

AJAY VENKATESH

Brooklyn, NY

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Education

New York University, New York, NY

Aug 2024 – May 2026

Master of Science in Computer Engineering (GPA: 3.9/4.0)

Coursework: System Design, Cloud Computing, Machine Learning, Big Data, Advanced Java, Data Structures

Experience

• CampusMesh

Jan 2026 – May 2026

Software Engineering Intern

New York, US

- Architected a **serverless backend** on AWS using **Serverless Framework**, **Lambda (Node.js 20)**, API Gateway, RDS/MySQL, DynamoDB, S3, and CloudFront; deployed across **4 environments** with SSM configs, KMS keys, and **least-privilege IAM**.
- Integrated **AWS Chime SDK** for in-space audio/video calls with idempotent meeting creation and a scheduled Lambda sweeper; modeled high-throughput chat in DynamoDB with partition+sort keys to eliminate hot partitions.
- Built a **JWT custom authorizer** with **AES-encrypted** user IDs and esbuild-bundled Lambdas to cut cold starts; designed a layered handler → service → repository → model architecture across **20+ endpoints** for the Study Space service.

• Amazon

May 2025 – Aug 2025

Software Development Engineer Intern

Seattle, US

- Designed a secure UI Console architecture using **Midway SSO**, **OAuth**, and **HELIS** to enforce least-privilege **role-based access (4+ roles)**; presented the solution to the team and received design approval.
- Provisioned **serverless Infrastructure-as-Code** using **AWS CDK** with API Gateway, Lambda, DynamoDB, S3, and IAM; built a **multi-stage, multi-region CI/CD pipeline** with automated promotion and rollback controls.
- Built a **React 18** frontend with optimized API integration, advanced filtering, and client-side caching, reducing page load time by **40%**.
- Engineered backend services in **Java** (Coral, Smithy, Dagger DI, CBOR); applied **low-level design patterns**, cutting backend execution time by **35%** with unit/integration testing via **Mockito**.

• New York University (Novel AI Technologies, Inc.)

Aug 2024 – Present

Software Developer

New York, US

- Architected an **NSF-funded (\$100K)** AI-powered health app for **iOS** and **Android** processing real-time wearable data; deployed cloud-hosted AI models for health metric analysis and food image processing.
- Engineered **ETL pipelines** transforming **NoSQL** into **PostgreSQL** with **concurrency control and locking**; processed **250K+** records and powered **AWS QuickSight** dashboards.
- Developed an insurance analytics platform (**React**, **Node.js**, **REST APIs**) deployed via **Docker** on **EC2**; implemented Stripe billing, RBAC, and row-level security, generating **\$50K+** revenue across **6 clients**.
- Streamlined underwriting via a broker portal integrated with insurer platforms; implemented **document/media processing** and **WebSocket-based real-time communication**.

• PricewaterhouseCoopers (PwC)

Aug 2021 – Aug 2024

Senior Software Engineer

Bengaluru, India

- Designed and implemented a **cloud-native distributed backend** on **Microsoft Azure** for a health platform, integrating **Microsoft Dynamics CRM** and event-driven Azure services for scalability and fault tolerance over **SQL Server**.
- Modernized a legacy onboarding system by building a scalable **B2B platform** with modular services; integrated data pipelines, email event tracking, and BI reporting, improving automation by **70%**.
- Deployed an **NLP-based OCR neural model** on Azure to automate structured data extraction, reducing processing time by **75%**.
- Optimized data ingestion with **parallel processing** on **100K-row** Excel feeds, cutting execution from **5 hours to 1 hour**; mentored engineers on **software design principles**, reducing production bugs by **25%**.

Projects

• InviLite | AWS, SageMaker, Bedrock, FAISS, Python

- Architected a cloud-native data pipeline (Kinesis, Lambda, S3, Glue, DynamoDB) ingesting structured and unstructured enterprise data into governed ML pipelines.
- Built a **RAG system** (SVM + FAISS + **Bedrock LLM**), cutting incident resolution from **1–2 hours to minutes** and lifting root-cause classification accuracy from **56% to 92%**.

• Flight Delay Prediction | PySpark, HDFS, Random Forest, Gradient Boosted Tree

- Built a scalable **PySpark ML pipeline** on HDFS to process and engineer features from over **7M flight records** using SQL-based aggregation and anomaly detection.
- Trained and optimized **Random Forest and Gradient Boosted Tree** models reaching **85%+** accuracy and converted outputs into actionable insights for data-driven scheduling.

Skills

- **Programming Languages:** C#, Python, Java, C++, JavaScript, TypeScript, SQL
- **Cloud (Azure):** Azure AI Services, Azure AI Foundry, Cognitive Services, Logic Apps, Function Apps, Storage Blobs, Data Lakes, Azure DevOps
- **Cloud (AWS):** CDK, SDK, EC2, S3, Lambda, IAM, SageMaker, Bedrock, API Gateway, DynamoDB
- **Databases:** SQL Server, Dataverse, PostgreSQL, MySQL, MongoDB, DynamoDB
- **Microsoft Stack:** .NET, Power Platform, Power Automate, Power BI, Microsoft Copilot Studio, Microsoft Dynamics CRM
- **DevOps & Tools:** Docker, Kubernetes, CI/CD Pipelines, PowerShell, Git, Visual Studio, Postman
- **AI/ML:** RAG, MLOps, AI Solution Deployment, Responsible AI, Model Validation & Monitoring

Publications

Supervised ML System Based Segmentation and Classification of Stroke Using Deep Learning Techniques (*IEEE*)

An Aquila-optimized SVM Classifier for Diabetes Prediction (*IEEE*)

Certifications

AWS Certified Cloud Practitioner | Microsoft Certified: Azure Data Scientist Associate | Microsoft Certified: Azure Data Fundamentals